

Practice questions: Vitamins and minerals

1. A 62-year-old woman comes to the physician because of a 1-month history of bone pain. She has a 5-year-history of chronic renal failure and is on regular dialysis treatment. Which of the following laboratory findings is most likely in this patient?

- A. Increased ALP and GGT levels
- B. Decreased serum calcium and phosphate levels
- C. Increased serum calcium and phosphate levels
- D. Increased serum calcitriol levels
- E. Decreased erythrocyte transketolase activity

2. A 2-year-old girl is brought to the physician by her mother because of a 1-month history of tripping over objects in the dark. Physical examination shows white spots on both the conjunctiva. A diagnosis of a nutritional deficiency disorder is made. Which of the following is the most likely risk factor for developing this deficiency?

- A. Administration of broad spectrum antibiotics
- B. Reduced bile entry into the intestine
- C. Chronic renal failure
- D. Hartnup disease
- E. Autoimmune parietal cell destruction

3. A 3-year-old boy is brought to the physician by his mother because of a 2-month history of bowing of the legs. Laboratory studies show elevated serum alkaline phosphatase levels and decreased serum calcium and decreased serum phosphate levels. A diagnosis of a vitamin deficiency is made. Which of the following is the most likely precursor of the deficient vitamin in the skin?

- A. 7-dehydrocholesterol
- B. Calcitriol
- C. 25-hydroxycholecalciferol
- D. Beta carotene
- E. Tryptophan

4. A 30-year-old woman comes to the physician because of a 2-month history of increased bruising and increased

bleeding tendency. She has a 3-year history of intermittent abdominal pain due to gall stones. A vitamin deficiency secondary to obstructive jaundice is confirmed. Which of the following is most likely an associated laboratory finding in this patient?

- A. Prolonged prothrombin time
- B. Elevated serum methylmalonate levels
- C. Elevated serum homocysteine levels
- D. Increased serum calcium and phosphate levels
- E. Decreased serum ferritin levels

5. A 30-year-old woman comes to the physician because of a 2-month history of increased bruising and increased bleeding tendency. She has a 3-year history of intermittent abdominal pain due to gall stones. A vitamin deficiency secondary to obstructive jaundice is confirmed. Which of the following best describes the normal function of this vitamin?

- A. Combines with opsin to form rhodopsin
- B. Facilitates calcium and phosphate absorption from the intestine
- C. Acts as a coenzyme for gamma-carboxylation of clotting factors
- D. Functions as an antioxidant and helps to detoxify free radicals
- E. Coenzyme for oxidation reduction reactions in metabolism

6. A 2-month-old boy is brought to the physician by his parents for a follow-up examination. He has had two episodes of seizures. Laboratory studies show elevated levels of branched chain keto acids in circulation. A diagnosis of an inherited enzyme deficiency disorder is made. A vitamin is supplemented to increase the function of the deficient enzyme. Which of the following vitamins is most likely supplemented to this patient?

- A. Biotin
- B. Tetrahydrobiopterin
- C. Tetrahydrofolate
- D. Calcitriol
- E. Thiamine

7. A 50-year-old woman comes to the physician because of a 4-week history of intermittent flushing and diarrhea. Laboratory studies show elevated levels of 5-HIAA in urine. Which of the following vitamin deficiencies is she most likely at risk for?

- A. Thiamine
- B. Niacin
- C. Vitamin D
- D. Vitamin K
- E. Cobalamin

8. A 32-year-old man comes to the physician because of a 1-month history of tingling and numbness of his hands and feet. He is on multidrug therapy including isoniazid for the management of tuberculosis for 2 months. A diagnosis of a vitamin deficiency is made and supplements of the vitamin are prescribed. Which of the following best describes the normal function of this vitamin coenzyme?

- A. Required for pyruvate dehydrogenase complex
- B. Coenzyme for methionine synthase
- C. Required for formation of dopamine from DOPA
- D. Required for gamma-carboxylation of prothrombin
- E. Required for ferrochelatase of heme synthesis

9. A 32-year-old man comes to the physician because of a 1-month history of tingling and numbness of his hands and feet. He is on multidrug therapy including isoniazid for the management of tuberculosis for 2 months. A diagnosis of a vitamin deficiency is made and supplements of the vitamin are prescribed. Which of the following best describes the normal function of this vitamin coenzyme?

- A. Required for fatty acid biosynthesis
- B. Required for condensation of succinyl CoA and glycine
- C. Component of ceruloplasmin
- D. Required for gluconeogenesis
- E. Required for the electron transport chain complexes

10. A 78-year-old woman is brought to the physician by her daughter because of a 1-month history of tingling and numbness of her hands and feet. She has also been depressed. Physical examination shows conjunctival pallor. Laboratory studies show elevated serum homocysteine and elevated methylmalonate levels. A diagnosis of a vitamin deficiency is made. Which of the following best describes the absorption characteristics of the vitamin deficient in this patient?

- A. Absorbed in stomach

- B. Absorption enhanced by dietary vitamin C
- C. Absorption takes place in terminal ileum
- D. Absorption requires bile salts
- E. Absorption requires HCl secreted by parietal cells

11. A 32-year-old woman comes to the physician because of a 2-month history of weakness. Physical examination shows conjunctival pallor. Laboratory studies show elevated serum homocysteine levels and serum methylmalonate levels are within normal limits. A diagnosis of a nutritional vitamin deficiency is made. Which of the following best describes the metabolism of this vitamin?

- A. Warfarin administration is a risk factor for developing deficiency
- B. Required for purine and thymidine synthesis
- C. Required for the conversion of propionyl CoA to succinyl CoA
- D. Absorption requires bile salts and chylomicrons
- E. Deficiency results in microcytic anemia

12. A 40-year-old man comes to the physician because of a 2-month history of mood changes and depression. Physical examination shows jaundice. Ophthalmoscopic examination shows a brownish ring around the cornea. Which of the following laboratory findings is most likely in this patient?

- A. Increased serum ceruloplasmin levels
- B. Increased serum ferritin levels
- C. Increased serum total iron binding capacity (TIBC)
- D. Increased urinary copper levels
- E. Increased urinary VMA levels

13. A 50-year-old man comes to the physician because of a 3-month history of tiredness and pain in the small joints of hands. Physical examination shows mild scleral icterus. Laboratory studies show a mild elevation in plasma glucose levels and elevated serum ferritin levels. Which of the following is most likely responsible for this clinical presentation?

- A. Reduced dietary copper absorption
- B. Defect in incorporation of iron to transferrin
- C. Increased dietary iron absorption
- D. Defect in the secretion of copper into the bile

E. Defect in myelination of neurons

Identify the types of anemia and lab tests in the following nutritional deficiency disorders:

	Type of anemia	Lab tests
Iron deficiency		
Pyridoxine deficiency		
Copper deficiency		
Vitamin B12 deficiency		
Folic acid deficiency		

Answer Key:

1. B

2. B

3. A

4. A

- 5. C
- 6. E
- 7. B
- 8. C
- 9. B
- 10. E
- 11. B
- 12. D
- 13. C

Short explanations for the practice questions:

1. A 62-year-old woman comes to the physician because of a 1-month history of bone pain. She has a 5-year-history of chronic renal failure and is on regular dialysis treatment. Which of the following laboratory findings is most likely in this patient?

Comment: This is a patient with vitamin D deficiency due to chronic renal failure. There is reduced formation of 1,25-DHCC due to low activity of 1-hydroxylase in the kidney

- A. Increased ALP and GGT levels – Seen in patients with obstructive jaundice/ cirrhosis (lower levels)
- B. Decreased serum calcium and phosphate levels – CORRECT ANSWER – 1,25-DHCC is required for the absorption of calcium and phosphate from the diet**
- C. Increased serum calcium and phosphate levels – This is seen in hypervitaminosis D
- D. Increased serum calcitriol levels – Serum calcitriol levels are low in chronic renal failure
- E. Decreased erythrocyte transketolase activity – seen in thiamine deficiency

2. A 2-year-old girl is brought to the physician by her mother because of a 1-month history of tripping over objects in the dark. Physical examination shows white spots on both the conjunctiva. A diagnosis of a nutritional deficiency disorder is made. Which of the following is the most likely risk factor for developing this deficiency?

Comment: This is a child with vitamin A deficiency. Vitamin A is a fat soluble vitamin and requires bile salts for its absorption. The child has night blindness and bitot spots.

- A. Administration of broad spectrum antibiotics – is a risk factor for vitamin K deficiency
- B. Reduced bile entry into the intestine – CORRECT ANSWER – is a risk factor for the development of fat-soluble vitamin (A, D or K) deficiency.**

C. Chronic renal failure – Is a risk factor for vitamin D deficiency

D. Hartnup disease – is a risk factor for niacin/NAD deficiency

E. Autoimmune parietal cell destruction – Is a risk factor for vitamin B12 deficiency. Parietal cells secrete intrinsic factor and HCl

3. A 3-year-old boy is brought to the physician by his mother because of a 2-month history of bowing of the legs. Laboratory studies show elevated serum alkaline phosphatase levels and decreased serum calcium and decreased serum phosphate levels. A diagnosis of a vitamin deficiency is made. Which of the following is the most likely precursor of the deficient vitamin in the skin?

Comment: This is a child with rickets

A. **7-dehydrocholesterol –CORRECT ANSWER – is converted to cholecalciferol (vitamin D) in the skin by the action of UV light. It is later hydroxylated to 1,25-DHCC in the liver and kidney.**

B. Calcitriol – is also known as 1,25 DHCC and is the active form of vitamin D

C. 25-hydroxycholecalciferol – is formed from vitamin D in the liver. Inactive form and has to be further activated by 1-hydroxylase in the kidney

D. Beta carotene – is the provitamin form of vitamin A

E. Tryptophan – is a precursor of NAD

4. A 30-year-old woman comes to the physician because of a 2-month history of increased bruising and increased bleeding tendency. She has a 3-year history of intermittent abdominal pain due to gall stones. A vitamin deficiency secondary to obstructive jaundice is confirmed. Which of the following is most likely an associated laboratory finding in this patient?

Comment: Bruising and increased bleeding tendency are signs of a bleeding disorder. Patients with obstructive jaundice are at risk for developing fat soluble vitamin deficiency (vitamin K)

A. **Prolonged prothrombin time – CORRECT ANSWER – Factor VII levels are the most sensitive to vitamin K deficiency**

B. Elevated serum methylmalonate levels – seen in vitamin B12 deficiency with elevated homocysteine and macrocytic anemia

C. Elevated serum homocysteine levels – seen in B12, folate and B6 deficiency. B12 and folate have associated macrocytic anemia; B6 deficiency is associated with microcytic anemia

D. Increased serum calcium and phosphate levels – seen in hypervitaminosis D

E. Decreased serum ferritin levels – seen in iron deficiency anemia

5. A 30-year-old woman comes to the physician because of a 2-month history of increased bruising and increased

bleeding tendency. She has a 3-year history of intermittent abdominal pain due to gall stones. A vitamin deficiency secondary to obstructive jaundice is confirmed. Which of the following best describes the normal function of this vitamin?

Comment: Bruising and increased bleeding tendency are signs of a bleeding disorder. Patients with obstructive jaundice are at risk for developing fat soluble vitamin deficiency (vitamin K)

A. Combines with opsin to form rhodopsin – this is the function of vitamin A (11-cis retinal)

B. Facilitates calcium and phosphate absorption from the intestine – this is the function of vitamin D

C. **Acts as a coenzyme for gamma-carboxylation of clotting factors – CORRECT ANSWER – Function of vitamin K**

D. Functions as an antioxidant and helps to detoxify free radicals – this is the function of vitamin E, selenium, vitamin C and beta-carotene

E. Coenzyme for oxidation reduction reactions in metabolism – Niacin (NAD/ NADP) and riboflavin (FAD, FMN) are coenzymes in oxidation-reduction reactions

6. A 2-month-old boy is brought to the physician by his parents for a follow-up examination. He has had two episodes of seizures. Laboratory studies show elevated levels of branched chain keto acids in circulation. A diagnosis of an inherited enzyme deficiency disorder is made. A vitamin is supplemented to increase the function of the deficient enzyme. Which of the following vitamins is most likely supplemented to this patient?

Comment: This is a child with maple syrup urine disease due to deficiency of branched chain alpha keto acid dehydrogenase

A. Biotin – Used in patients with multiple carboxylase deficiency; Pyruvate carboxylase, acetyl CoA carboxylase and Propionyl CoA carboxylase are dependent on biotin.

B. Tetrahydrobiopterin – Used in PKU-II and in PKU-I; Tetrahydrobiopterin is required for conversion of phenylalanine to tyrosine, tyrosine to DOPA and tryptophan to 5-hydroxy tryptophan

C. Tetrahydrofolate – Used for purine and thymidine synthesis (DNA synthesis) and for methionine synthase (homocysteine methyl transferase) reactions

D. Calcitriol – Required for optimal absorption of dietary calcium and phosphate

E. **Thiamine – CORRECT ANSWER – TPP is required as a coenzyme for branched chain alpha keto acid dehydrogenase and has been tried in patients with MSUD**

7. A 50-year-old woman comes to the physician because of a 4-week history of intermittent flushing and diarrhea. Laboratory studies show elevated levels of 5-HIAA in urine. Which of the following vitamin deficiencies is she most likely at risk for?

Comment: This is a patient with carcinoid syndrome secreting serotonin (derived from tryptophan). Patients with carcinoid tumors are at risk of developing pellagra (niacin deficiency)

A. Thiamine - Chronic alcoholics are at risk for Wernicke Korsakoff syndrome

B. **Niacin – CORRECT ANSWER - NAD can be formed from tryptophan**

C. Vitamin D – Patients with liver disease, chronic renal failure and reduced exposure to sunlight are at risk for vitamin D deficiency

D. Vitamin K – Patients with obstructive jaundice and patients on oral broad spectrum antibiotics are at risk

E. Cobalamin – Patients with parietal cell destruction and achlorhydria are at risk. Remember, intrinsic factor and HCl are secreted by parietal cells. Patients with terminal ileal disease or ileal resection are also at risk

8. A 32-year-old man comes to the physician because of a 1-month history of tingling and numbness of his hands and feet. He is on multidrug therapy including isoniazid for the management of tuberculosis for 2 months. A diagnosis of a vitamin deficiency is made and supplements of the vitamin are prescribed. Which of the following best describes the normal function of this vitamin coenzyme?

Comment: Patients on isoniazid therapy are at risk for B6 (pyridoxine) deficiency

A. Required for pyruvate dehydrogenase complex – PDH complex requires TPP, NAD, FAD, CoA and lipoic acid as coenzymes

B. Coenzyme for methionine synthase – Also known as homocysteine methyl transferase; Requires methyl tetrahydrofolate and B12 as coenzymes

C. **Required for formation of Dopamine from DOPA – CORRECT ANSWER** – PLP is required as a coenzyme for amino acid decarboxylation reactions (neurotransmitter synthesis); aminotransferase reactions and cystathionine synthase (deficient in inherited homocystinuria) and heme synthesis (delta-ALA synthase)

D. Required for gamma-carboxylation of prothrombin – this is the function of vitamin K

E. Required for ferrochelatase of heme synthesis – this step of heme synthesis requires iron; It is inhibited by lead

9. A 32-year-old man comes to the physician because of a 1-month history of tingling and numbness of his hands and feet. He is on multidrug therapy including isoniazid for the management of tuberculosis for 2 months. A diagnosis of a vitamin deficiency is made and supplements of the vitamin are prescribed. Which of the following best describes the normal function of this vitamin coenzyme?

Comment: Patients on isoniazid therapy are at risk for B6 (pyridoxine) deficiency; Slightly different from the previous question

A. Required for fatty acid biosynthesis – Biotin is a coenzyme for acetyl CoA carboxylase; and NADPH is required for fatty acid biosynthesis

B. **Required for condensation of succinyl CoA and glycine – CORRECT ANSWER** – PLP is required for delta ALA synthase reaction in the heme biosynthesis pathway

C. Component of ceruloplasmin – copper is a component of ceruloplasmin

D. Required for gluconeogenesis – Biotin is a coenzyme for pyruvate carboxylase

E. Required for the electron transport chain complexes – FMN, FAD, iron and copper are components of the ETC complexes

10. A 78-year-old woman is brought to the physician by her daughter because of a 1-month history of tingling and numbness of her hands and feet. She has also been depressed. Physical examination shows conjunctival pallor. Laboratory studies show elevated serum homocysteine and elevated methylmalonate levels. A diagnosis of a vitamin deficiency is made. Which of the following best describes the absorption characteristics of the vitamin deficient in this patient?

Comment: This is a patient with vitamin B12 deficiency. Note the anemia (macrocytic) with neurological features. Serum homocysteine and methylmalonate levels are elevated!

A. Absorbed in stomach – Parietal cells secrete intrinsic factor which bind to dietary vitamin B12.

B. Absorption enhanced by dietary vitamin C – Iron absorption is enhanced by dietary vitamin C. Iron deficiency is characterized by microcytic anemia and NO neurological features.

C. **Absorption takes place in terminal ileum – CORRECT ANSWER – IF-B12 complex is absorbed in terminal ileum**

D. Absorption requires bile salts – This is true for the fat soluble vitamins

E. Absorption requires HCl secreted by parietal cells – Parietal cells also secrete intrinsic factor which is required for the absorption of vitamin B12; HCl is not required for B12 absorption.

11. A 32-year-old woman comes to the physician because of a 2-month history of weakness. Physical examination shows conjunctival pallor. Laboratory studies show elevated serum homocysteine levels and serum methylmalonate levels are within normal limits. A diagnosis of a nutritional vitamin deficiency is made. Which of the following best describes the metabolism of this vitamin?

Comment: The patient has folate deficiency. Note the anemia (macrocytic); no neurological features. Elevated serum homocysteine and normal methylmalonate levels.

A. Warfarin administration is a risk factor for developing deficiency – Methotrexate administration is a risk factor for folate deficiency; Warfarin administration is a risk for vitamin K deficiency

B. **Required for purine and thymidine synthesis – CORRECT ANSWER – Tetrahydrofolate is required for DNA synthesis. Patients with folate deficiency develop macrocytic anemia**

C. Required for the conversion of propionyl CoA to succinyl CoA – Biotin and vitamin B12 are required for the conversion (Propionyl CoA carboxylase and methylmalonyl CoA mutase)

D. Absorption requires bile salts and chylomicrons – This is a water soluble vitamin and does not require bile salts

E. Deficiency results in microcytic anemia – Folate deficiency results in macrocytic anemia

12. A 40-year-old man comes to the physician because of a 2-month history of mood changes and depression. Physical examination shows jaundice. Ophthalmoscopic examination shows a brownish ring around the cornea. Which of the following laboratory findings is most likely in this patient?

Comment: This is a patient with Wilson disease – note the neurological changes (basal ganglia), jaundice and Kayser-

Fleischer ring in cornea

A. Increased serum ceruloplasmin levels – Serum ceruloplasmin levels are low

B. Increased serum ferritin levels – Seen in hemochromatosis

C. Increased serum total iron binding capacity (TIBC) – Seen in iron deficiency anemia

D. **Increased urinary copper levels – CORRECT ANSWER** – Lab tests useful in diagnosis of Wilson disease

E. Increased urinary VMA levels – Lab tests for pheochromocytoma

13. A 50-year-old man comes to the physician because of a 3-month history of tiredness and pain in the small joints of hands. Physical examination shows mild scleral icterus. Laboratory studies show a mild elevation in plasma glucose levels and elevated serum ferritin levels. Which of the following is most likely responsible for this clinical presentation?

Comment: This is a patient with hemochromatosis – note the involvement of liver (icterus), hyperglycemia (iron deposition in beta cells of pancreas) and increased serum ferritin levels. Also known as “Bronze diabetes”

A. Reduced dietary copper absorption – this is seen in Menkes disease

B. Defect in incorporation of iron to transferrin – Incorporation of iron into transferrin is done by ceruloplasmin which functions as a ferroxidase

C. **Increased dietary iron absorption – CORRECT ANSWER** – due to defect in regulation of iron absorption; there is a mutation of the HFE gene

D. Defect in the secretion of copper into the bile – this is observed in Wilson disease

E. Defect in myelination of neurons – this is observed in vitamin B12 deficiency

Identify the types of anemia and lab tests in the following nutritional deficiency disorders:

	Type of anemia	Lab tests
Iron deficiency	Microcytic hypochromic anemia	Low MCV, Low MCHC; Low serum ferritin; Low serum iron; Increased TIBC; Decreased %Transferrin saturation; Low Bone marrow iron stores
Pyridoxine deficiency	Microcytic anemia	Low MCV; Elevated Homocysteine; Iron parameters - Normal
Copper deficiency	Microcytic anemia	Low MCV; Low serum copper

Vitamin B12 deficiency	Macrocytic, megaloblastic anemia; Neuropsychiatric manifestations	MCV – Increased; Increased Homocysteine; Increased methylmalonate; Low Vitamin B12 levels
Folic acid deficiency	Macrocytic, megaloblastic anemia	MCV – Increased; Increased Homocysteine; Methylmalonate - Normal; Low Folic acid levels; Normal vitamin B12 levels

Practice Questions: Obesity

1. A 25-year-old man comes to the physician for a routine examination. His BMI is 30 kg/m². Laboratory studies show high levels of a specific hormone which acts as a long-term appetite suppressant. Which of the following hormones is most likely elevated in this patient?
 - A. Glucagon
 - B. Insulin
 - C. Ghrelin
 - D. Adiponectin
 - E. Cortisol
2. A 45-year-old woman comes to the physician for a routine examination. Her BMI is 32kg/m². Clinical evaluation and laboratory studies confirm insulin resistance syndrome. Which of the following laboratory findings is most likely in this patient?
 - A. Low levels of LDL cholesterol
 - B. Low fasting plasma glucose level
 - C. High levels of circulating insulin
 - D. Low levels of TAGs
 - E. Low levels of free fatty acids
3. A 35-year-old woman comes to the physician for a follow-up examination. Laboratory studies show elevated circulating ghrelin levels in circulation due to overstimulation of the *ghrelin* gene. Which of the following is most likely in this patient?

- A. Weight gain due to increased appetite
 - B. Weight loss due to suppressed appetite
 - C. Weight gain due to increased insulin levels
 - D. Weight gain due to increased adiponectin
 - E. Increased resting energy expenditure and weight loss
4. A 45-year-old man comes to the physician for a routine examination. His BMI is 31.5kg/m² and waist to hip ratio (WHR) is 1.3. A diagnosis of insulin resistance syndrome is made. Which of the following laboratory findings is most likely in this patient?
- A. Normal levels of circulating insulin
 - B. Low levels of circulating insulin
 - C. High levels of VLDL
 - D. Low levels of TAGs
 - E. High level of HDL cholesterol
5. A 30-year-old woman comes to the physician for a follow-up examination. Her BMI is 30.7kg/m² and waist circumference is 39 inches. Laboratory studies show elevated free fatty acid levels. Which of the following is most likely responsible for this laboratory finding?
- A. Increased hormone sensitive lipase activity
 - B. Increased activity of phosphoenolpyruvate carboxy kinase (PEPCK)
 - C. Decreased activity of lipoprotein lipase
 - D. Decreased uptake of LDL by the peripheral tissues
 - E. Increased fatty acid synthesis stimulated by leptin
6. A 25-year-old man comes to the physician for weight management. His triceps skin-fold thickness is obtained to assess the subcutaneous fat content. Which of the following measurements best assesses this parameter?
- A. Bioelectric impedance
 - B. Body mass index (BMI)
 - C. Height measurement
 - D. Weight measurement
 - E. Waist circumference measurement

7. A 55-year-old woman comes to the physician for a follow-up examination. She is a type 2 diabetic for 5 years. Her blood pressure is slightly elevated. Laboratory studies show elevated fasting serum TAG. Which of the following best explains this laboratory finding?
- A. Decreased hormone sensitive lipase activity
 - B. Increased PEPCK activity in liver
 - C. Reduced number of GLUT-4 in adipose tissue
 - D. Decreased lipoprotein lipase activity
 - E. Shift of water from ICF to ECF

Answer Key

- 1 = B
- 2 = C
- 3 = A
- 4 = C
- 5 = A
- 6 = A
- 7 = D

Short explanations for Practice questions on Obesity:

1. A 25-year-old man comes to the physician for a routine examination. His BMI is 30 kg/m². Laboratory studies show high levels of a specific hormone which acts as a long-term appetite suppressant. Which of the following hormones is most likely elevated in this patient?

- A. Glucagon
- B. **Insulin – Correct answer:** Obese individuals tend to have less insulin responsiveness in the target tissues (insulin resistance) and hence increased insulin secretion; Leptin levels are also increased in obese patients

- C. Ghrelin – levels are elevated in the fasting state and it stimulates hunger
- D. Adiponectin – levels are low in obese individuals. Low adiponectin secretion contributes to the development of insulin resistance
- E. Cortisol – levels are elevated in stress and Cushing syndrome

2. A 45-year-old woman comes to the physician for a routine examination. Her BMI is 32kg/m². Clinical evaluation and laboratory studies confirm insulin resistance syndrome. Which of the following laboratory findings is most likely in this patient?

- A. Low levels of LDL cholesterol – Insulin resistance usually results in dyslipidemia and elevated LDL cholesterol levels (atherogenic profile)
- B. Low fasting plasma glucose level – Insulin resistance usually results in decreased glucose tolerance and elevated plasma glucose levels
- C. **High levels of circulating insulin – Correct answer: Obese individuals tend to have less insulin responsiveness in the target tissues (insulin resistance) and hence increased insulin secretion**
- D. Low levels of TAGs – Serum TAG is usually elevated due to low activity of lipoprotein lipase. Remember that LPL requires insulin for optimum activity. Low LPL activity results in increased TAG in VLDL. TAG in VLDL is not optimally cleaved to fatty acids in endothelium of adipocytes and serum TAG is elevated.
- E. Low levels of free fatty acids – Free fatty acids in circulation is due to the activity of adipocyte hormone sensitive lipase. In obesity and insulin resistance, HS lipase is increased which in turn results in increased free fatty acids in circulation. Remember, HS-Lipase is inhibited by insulin. Increased serum free fatty acids is one of the laboratory findings in patients with insulin resistance syndrome (metabolic syndrome)

3. A 35-year-old woman comes to the physician for a follow-up examination. Laboratory studies show elevated circulating ghrelin levels in circulation due to overstimulation of the *ghrelin* gene. Which of the following is most likely in this patient?

- A. **Weight gain due to increased appetite – Correct answer: Ghrelin secreted from the stomach stimulates hunger and increases food intake**
- B. Weight loss due to suppressed appetite – Leptin levels are increased when the adipocytes increase in size and a person gains weight. High leptin levels suppress appetite; but it is observed that most obese patients have leptin resistance and leptin does not function optimally.
- C. Weight gain due to increased insulin levels – Weight gain results in increased insulin levels due to insulin resistance;

- D. Weight gain due to increased adiponectin – Adiponectin secretion from adipocytes is lower in obesity
- E. Increased resting energy expenditure and weight loss – Leptin increases energy expenditure

4. A 45-year-old man comes to the physician for a routine examination. His BMI is 31.5kg/m² and waist to hip ratio (WHR) is 1.3. A diagnosis of insulin resistance syndrome is made. Which of the following laboratory findings is most likely in this patient?

- A. Normal levels of circulating insulin – Insulin levels would most likely be elevated due to target tissue resistance to insulin
- B. Low levels of circulating insulin – Insulin levels would most likely be elevated
- C. **High levels of VLDL – Correct answer:** Insulin resistance results in low activity of lipoprotein lipase resulting increased VLDL levels and increased serum TAGs
- D. Low levels of TAGs – Serum TAGs levels are elevated due to increased VLDL levels
- E. High level of HDL cholesterol – HDL levels are low in insulin resistance and increases the risk of atherosclerosis and cardiovascular disease

5. A 30-year-old woman comes to the physician for a follow-up examination. Her BMI is 30.7kg/m² and waist circumference is 39 inches. Laboratory studies show elevated free fatty acid levels. Which of the following is most likely responsible for this laboratory finding?

- A. **Increased hormone sensitive lipase activity – Correct answer:** Elevated free fatty acids is due to increased adipocyte hormone sensitive lipase activity. HS lipase is generally inhibited by insulin. In insulin resistance, there is increased HS lipase activity (Increased lipolysis in adipose tissue) which leads to increased free fatty acid levels in circulation.
- B. Increased activity of phosphoenolpyruvate carboxy kinase (PEPCK) – responsible for the hyperglycemia. The PEPCK in liver is more active resulting in increased gluconeogenesis and increased plasma glucose levels
- C. Decreased activity of lipoprotein lipase – is observed in insulin resistance syndrome. This leads to increased VLDL levels and increased serum TAG
- D. Decreased uptake of LDL by the peripheral tissues – LDL levels are usually elevated leading to a higher risk of atherosclerosis in obese individuals
- E. Increased fatty acid synthesis stimulated by leptin – Leptin levels are elevated in obese patients; But is not responsible for the elevated free fatty acids

6. A 25-year-old man comes to the physician for weight management. His triceps skin-fold thickness is obtained to assess the subcutaneous fat content. Which of the following measurements best assesses this parameter?

- A. Bioelectric impedance – **Correct answer**: Indicate the amount of adipose tissue and muscle mass ratio
- B. Body mass index (BMI) – estimates the weight to height in kg/m^2 . BMI measurements do not indicate the extent of subcutaneous fat/ muscle mass ratio.
- C. Height measurement – BMI is a better indicator than just the weight or height
- D. Weight measurement – BMI is a better indicator than just the weight or height
- E. Waist circumference measurement – Waist circumference is a good indicator of abdominal obesity. The waist to hip ratio (WHR) gives a good indication of subcutaneous fat to visceral fat. Higher the WHR, greater is the visceral fat (central obesity/ abdominal obesity) and higher is the risk of insulin resistance syndrome. The WHR is a better indicator than waist circumference.

7. A 55-year-old woman comes to the physician for a follow-up examination. She is a type 2 diabetic for 5 years. Her blood pressure is slightly elevated. Laboratory studies show elevated fasting serum TAG. Which of the following best explains this laboratory finding?

- A. Decreased hormone sensitive lipase activity – results in elevated free fatty acids
- B. Increased PEPCK activity in liver – Is responsible for increased gluconeogenesis in liver which results in hyperglycemia
- C. Reduced number of GLUT-4 in adipose tissue – Is responsible for hyperglycemia
- D. Decreased lipoprotein lipase activity – **Correct answer**; Elevated TAG is due to low activity of lipoprotein lipase.
- E. Shift of water from ICF to ECF – Is responsible for neurological changes in acute hyperglycemia

Practice MCQ for **Feed-Fast cycle**, **explanations**

1. Metabolic changes are observed in 5-year-old boys following a breakfast of cereal, milk and a big serving of honey. Metabolic changes are observed two hours later. Which of the following pathways will be active in their skeletal muscle during the period of observation?

- A. Pentose phosphate pathway
- B. Ketone body usage
- C. Protein synthesis
- D. Glycogen degradation
- E. Branched-chain amino acid synthesis

2. Metabolic changes are observed in a 5-year-old girl who eats breakfast at home which contains her favorite cereal, milk and a big serving of honey. Metabolism in adipose tissue is observed two hours later. Which of the following is most likely activated by insulin in this tissue?

- A. Hormone-sensitive lipase
- B. GLUT-1 transporters
- C. MAG pathway for TAG synthesis
- D. Lipoprotein lipase
- E. Fatty acid de novo synthesis

3. A research group is studying the role of hormones in the regulation of plasma glucose levels. It is observed that release of epinephrine supports glucagon action as both act via the cAMP messenger pathway. cAMP activates protein kinase A (PKA) which phosphorylates key regulatory enzymes and indirectly activates gluconeogenesis. Which of the following enzymes is most likely inhibited by phosphorylation by PKA to indirectly support gluconeogenesis?

- A. Pyruvate kinase
- B. Glycogen phosphorylase kinase
- C. Pyruvate dehydrogenase
- D. Acetyl CoA carboxylase
- E. FBPase-2 activity of the bifunctional enzyme

4. A 28-year-old man comes to the physician because he wants to lose weight. He participates in a study where laboratory data are collected for up to 10 days of voluntary fasting. Which of the following laboratory findings is most likely observed in this patient at the end of the fast?

- A. Blood glucose levels of 2.8 mM
- B. Higher levels of acetoacetate than 3-hydroxybutyrate
- C. Higher levels of free fatty acids than glucose
- D. Absent insulin levels
- E. Higher levels of 3-hydroxybutyrate than free fatty acids

5. A research group is studying the role of hormones on glycogen metabolism and gluconeogenesis in the liver. It is observed that cortisol is involved in regulation of plasma glucose levels. Which of the following best explains the action of this hormone?

- A. Activation of PEPcarboxykinase (PEPCK) by enzyme induction
- B. Inhibition of glycolysis by inhibition of pyruvate kinase
- C. Stimulation of glucagon release
- D. Inhibition of insulin release

E. Activation of protein synthesis in muscle

Answer Key: 1 C, 2 D, 3 A, 4 E, 5 A

Explanations for practice MCQ for Feed-Fast Cycle: role of hormones

1. Metabolic changes are observed in 5-year-old boys following a breakfast of cereal, milk and a big serving of honey. Metabolic changes are observed two hours later. Which of the following pathways will be active in their skeletal muscle during the period of observation?

- A. Pentose phosphate pathway **no**: this pathway has always a low activity in muscle
- B. Ketone body usage **no**: ketone bodies are normally not in the blood after a meal
- C. **Protein synthesis Correct! Insulin stimulates protein synthesis in the resting muscle**
- D. Glycogen degradation **no**: glycogen *synthesis* is stimulated by insulin
- E. Branched-chain amino acid synthesis **no**: they are dietary essential. In muscle, they can be used for protein synthesis or for energy metabolism

2. Metabolic changes are observed in a 5-year-old girl who eats breakfast at home which contains her favorite cereal, milk and a big serving of honey. Metabolism in adipose tissue is observed two hours later. Which of the following is most likely activated by insulin in this tissue?

- A. Hormone-sensitive lipase **no**: insulin leads to the dephosphorylated and inactive HSL
- B. GLUT-1 transporters **no**: GLUT-4 are found and activated by insulin
- C. MAG pathway for TAG synthesis **no**: this is performed in intestinal mucosal cells
- D. **Lipoprotein lipase Correct! Insulin stimulates the synthesis of LPL in fat cells and then anchoring of LPL in fat tissue capillaries. LPL cleaves TAG in chylomicrons to release fatty acids which enter into the adipocyte and are stored as TAG**
- E. Fatty acid de novo synthesis **no**: the fatty acids formed by LPL are taken up into fat cells for TAG synthesis. Blood glucose is used for the glycerol 3-P formation to provide the glycerol backbone in TAGs. In humans fatty acid synthesis is mainly performed in the liver using dietary glucose stimulated by insulin.

3. A research group is studying the role of hormones in the regulation of plasma glucose levels. It is observed that release of epinephrine supports glucagon action as both act via the cAMP messenger pathway. cAMP activates protein kinase A (PKA) which phosphorylates key regulatory enzymes and indirectly activates gluconeogenesis. Which of the following enzymes is most likely inhibited by phosphorylation by PKA to indirectly support gluconeogenesis?

- A. **Pyruvate kinase Correct! Glycolysis is inhibited and PEP can be used for gluconeogenesis.**
- B. Glycogen phosphorylase kinase **no**: it is not inhibited but activated by PKA
- C. Pyruvate dehydrogenase **no**: PDH is phosphorylated by PDH kinase
- D. Acetyl CoA carboxylase **no**: this enzyme is phosphorylated by AMP-kinase
- E. FBPase-2 activity of the bifunctional enzyme **no**: phosphorylation of the bifunctional

enzyme leads to an active FBPase-2. It is the PFK-2 activity that is inhibited when the bifunctional enzyme is phosphorylated.

4. A 28-year-old man comes to the physician because he wants to lose weight. He participates in a study where laboratory data are collected for up to 10 days of voluntary fasting. Which of the following laboratory findings is most likely observed in this patient at the end of the fast?

- A. Blood glucose levels of 2.8 mM **no**: Plasma glucose levels generally do not drop lower than 3.8 mM.
- B. Higher levels of acetoacetate than 3-hydroxybutyrate **no**: the other way round, higher levels of 3-hydroxybutyrate which is also more stable than acetoacetate.
- C. Higher levels of free fatty acids than glucose **no**: free fatty acid levels are elevated but not higher than blood glucose.
- D. Absent insulin levels **no**: both insulin and glucagon are in the blood. They allow fine-tuning of fasting blood glucose levels. Plasma glucose level will be maintained at around 3.8 mM.
- E. Higher levels of 3-hydroxybutyrate than free fatty acids **Correct!** 3-hydroxybutyrate levels increase in blood sharply up to 20 days of fasting. The fatty acids released by fat cells are taken up into the liver and used for synthesis of 3-hydroxybutyrate (ketone bodies).

5. A research group is studying the role of hormones on glycogen metabolism and gluconeogenesis in the liver. It is observed that cortisol is involved in regulation of plasma glucose levels. Which of the following best explains the action of this hormone?

- A. Activation of PEPcarboxykinase (PEPCK) by enzyme induction Correct! Cortisol is a steroid hormone that acts via protein synthesis and the synthesis of PEPcarboxykinase activates gluconeogenesis and is needed long term.**
- B. Inhibition of glycolysis by inhibition of pyruvate kinase **no**: that is performed by PKA following glucagon and epinephrine action.
- C. Stimulation of glucagon release **no**: epinephrine stimulates glucagon release.
- D. Inhibition of insulin release **no**: epinephrine inhibits insulin release.
- E. Activation of protein synthesis in muscle **no**: cortisol activates protein degradation in muscle

Practice questions: Lysosomal storage disorders

1. A 32-year-old man comes to the physician because of a 2-week history of pain in the right leg. An x-ray of the legs shows that the distal femur is shaped like an 'Erlenmeyer flask'. A peripheral blood smear and skin biopsy is performed. Which of the following types of cells is most likely to be observed?

- A. Inclusion bodies in the cytoplasm

- B. Crumpled tissue paper appearance of cytoplasm
- C. Foamy cell appearance of cytoplasm
- D. Onion shell inclusions in lysosomes
- E. Multi-nucleated macrophages

2. A 10-day-old girl dies due to cardiac failure. She has had poor feeding and failure to thrive since birth. An autopsy shows an accumulation of glycogen in the heart muscle. Biochemical tests are performed from the heart tissue. Which of the following enzymes is most likely deficient in this tissue?

- A. Aryl sulfatase A
- B. Hexosaminidase A
- C. β -glucosidase
- D. Acid maltase
- E. α -galactosidase

3. A 30-year-old woman comes to the physician for prenatal counseling. She has had a sister who had a genetic disorder and died when she was 4 years old. Blood samples are obtained and leukocytes are isolated. Biochemical tests are performed on the leukocytes. Laboratory studies show 50% activity of hexosaminidase A. This patient is most likely a carrier of which of the following lysosomal storage disorders?

- A. Gaucher
- B. Hunter
- C. Hurler
- D. Fabry
- E. Tay-Sachs

4. A 2-year old girl is brought to the pediatrician by her parents because of a delay in developmental milestones. The parents say that does not seem to be achieving developmental milestones as fast as her older sister did. Physical

examination shows facial dysmorphism with frontal bossing and slight clouding of the cornea. Which of the following lysosomal storage diseases is most likely in this patient?

- A. Hurler
- B. Hunter
- C. Tay-Sachs
- D. I-cell
- E. Pompe

5. Investigators in a biochemical genetics laboratory are performing urinalysis on samples from children with hepatosplenomegaly due to lysosomal storage diseases. Many lysosomal diseases lead to the accumulation of glycosaminoglycans in lysosomes which are excreted into urine. Which of the following diseases will most likely result in a positive glycosaminoglycans test? (choose as many as believe to be true)

- A. Hunter
- B. Hurler
- C. Tay-Sachs
- D. Pompe
- E. Metachromatic leukodystrophy
- F. I-cell
- G. Niemann-Pick

ANSWER KEY:

- 1. B
- 2. D
- 3. E
- 4. A
- 5. A, B, F

Answers to the Practice questions

1. A 32-year old man comes to the physician because of a 2-week history of pain in the right leg. An x-ray of the legs shows that the distal femur is shaped like an 'Erlenmeyer flask'. A peripheral blood smear and skin biopsy is performed. Which of the following types of cells is most likely to be observed?

A. Inclusion bodies in the cytoplasm – feature of I-cell disease; Defect in sorting enzymes to the lysosomes due to defective phosphorylation of mannose residues.

B. **Crumpled tissue paper appearance of cytoplasm – Correct answer:** Observed in Gaucher disease; Deficient enzyme is beta-glucosidase; Glucocerebrosides accumulate in lysosomes.

C. Foamy cell appearance of cytoplasm – Niemann-Pick disease; Deficiency of sphingomyelinase

D. Onion shell inclusion in lysosomes – Tay-Sachs disease; Children have developmental delay and blindness

E. Multi nucleated macrophages

2. A 10-day-old girl dies due to cardiac failure. She has had poor feeding and failure to thrive since birth. An autopsy shows an accumulation of glycogen in the heart muscle. Biochemical tests are performed from the heart tissue. Which of the following enzymes is most likely deficient in this tissue?

A. Aryl sulfatase A – Deficient in metachromatic leukodystrophy; Sulfatides accumulate

B. Hexosaminidase A – deficient in Tay-Sachs disease; Gangliosides accumulate

C. β -glucosidase – deficient in Gaucher disease; Glucocerebrosides accumulate

D. **Acid maltase – Correct answer:** The child has Pompe disease (glycogen storage disease type II). Cardiomegaly is a feature.

E. α -galactosidase – Deficient in Fabry disease; Globosides accumulate

3. A 30-year-old woman comes to the physician for prenatal counseling. She has had a sister who had a genetic disorder and died when she was 4 years old. Blood samples are obtained and leukocytes are isolated. Biochemical tests are performed on the leukocytes. Laboratory studies show 50% activity of hexosaminidase A. This patient is most likely a carrier of which of the following lysosomal storage disorders?

A. Gaucher – Beta-glucosidase is deficient

B. Hunter – Deficiency of iduronate sulfatase

C. Hurler – Deficiency of iduronidase

D. Fabry – Alpha-galactosidase deficiency

E. **Tay-Sachs – Correct answer:** Carriers have 50% activity of the enzyme and are phenotypically normal

4. A 2-year old girl is brought to the pediatrician by her parents because of a delay in developmental milestones. The parents say that she does not seem to be achieving developmental milestones as fast as her older sister did. Physical examination shows facial dysmorphism with frontal bossing and slight clouding of the cornea. Which of the following lysosomal storage diseases is most likely in this patient?

A. **Hurler – Correct answer: Accumulation of heparan and dermatan sulfate. Facial dysmorphism with hepatosplenomegaly and corneal clouding. Deficiency of iduronidase**

B. Hunter – X-linked disorder; Corneas are clear; Accumulation of heparan and dermatan sulfate. Iduronate sulfatase deficiency

C. Tay-Sachs disease – Autosomal recessive; deficiency of hexosaminidase A; Accumulation of gangliosides

D. I-cell – Defect in trafficking lysosomal enzymes due to a defect in mannose phosphorylation. Generalized defect in lysosomal degradation of all macromolecules (glycolipids and glycosaminoglycans).

E. Pompe – Defect in acid maltase and accumulation of glycogen in lysosomes

5. Investigators in a biochemical genetics laboratory are performing urinalysis on samples from children with hepatosplenomegaly due to lysosomal storage diseases. Many lysosomal diseases lead to the accumulation of glycosaminoglycans in lysosomes which are excreted into urine. Which of the following diseases will most likely result in a positive glycosaminoglycans test? (choose as many as believe to be true)

A. **Hunter – Accumulation of heparan and dermatan sulfate (GAG)**

B. **Hurler – Accumulation of heparan and dermatan sulfate (GAG)**

C. Tay-Sachs – Ganglioside (glycolipids) accumulation

D. Pompe – Lysosomal glycogen accumulation

E. Metachromatic leukodystrophy – Accumulation of sulfatides

F. **I-cell – Generalized deficiency of all lysosomal enzymes; Accumulation of GAGs, glycolipids, phospholipids**

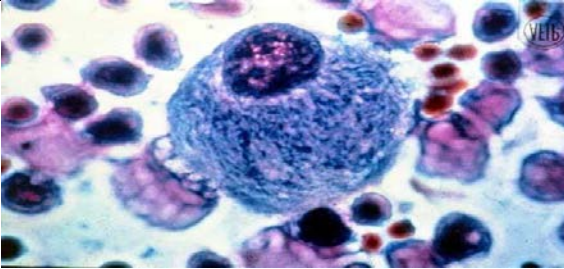
G. Niemann-Pick - Accumulation of sphingomyelin; Deficiency of sphingomyelin (sphingophospholipid)

Lysosomal Storage Disorders

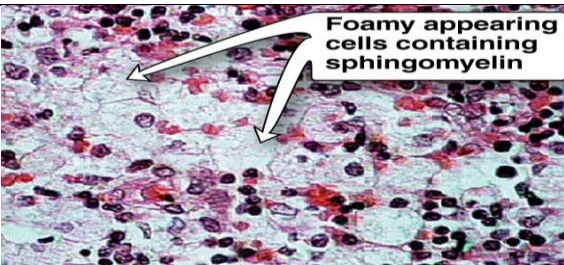
Match each histological finding with its Lysosomal Storage Disorder: Identify the compounds accumulating in each of the disorders



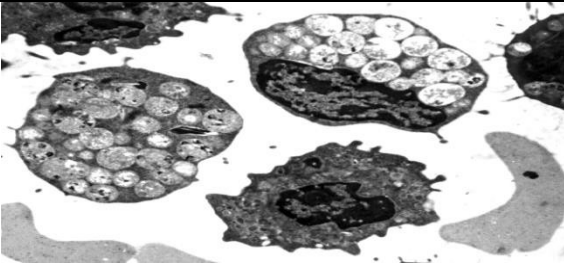
Niemann-Pick



I-Cell Disease



Gaucher Disease

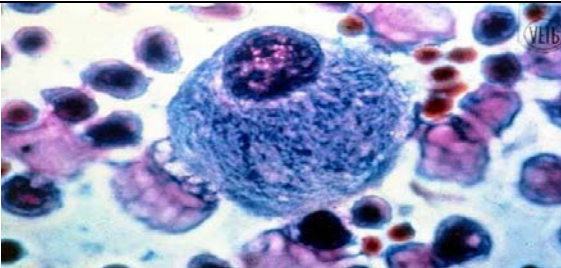
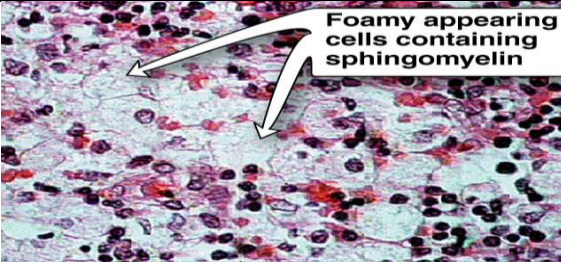
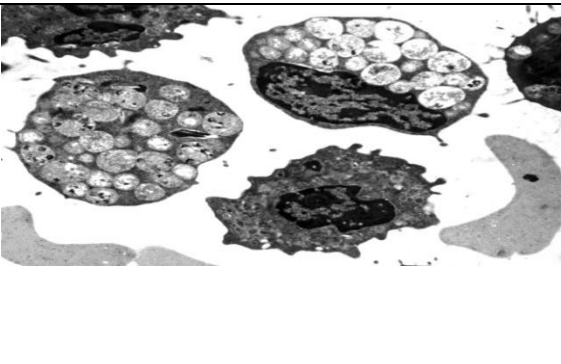


Tay-Sachs Disease

Complete the following table:

Disorder	Defective Enzyme	Accumulating substrate	Clinical Manifestations	Histological findings
Hurler Syndrome AR		Heparan Sulfate, Dermatan Sulfate	Corneal clouding, dysmorphic facial features, hepatosplenomegaly, joint contractures	
	Iduronate sulfatase	Heparan Sulfate, Dermatan Sulfate	NO Corneal clouding, dysmorphic facial features, hepatosplenomegaly, Joint contractures, "Mongolian" spot on lower back	
Tay- Sachs Disease AR	Hexosaminidase A		Cherry-red spot on retina (blindness), neurological and developmental regression. No hepato-splenomegaly	
Gaucher Disease AR			Hepatosplenomegaly, bone pain. Sparing the brain	Crumpled paper appearance
Fabry Disease XR			Numbness & tingling in extremities (acroparesthesia); red spots on buttocks and thighs (angiokeratoma) heart, kidney dysfunction	
	Sphingomyelinase	Sphingomyelin	Type A- cherry red spot on retina. Type B- hepatosplenomegaly	
Metachromatic Leukodystrophy AR			Progressive paralysis, motor & sensory abnormalities	
Pompe Disease (GSD type II) AR	Lysosomal acid maltase (1-4 glucosidase)		Cardiomyopathy, Muscle weakness, CVS, liver, kidney dysfunction	
	N-acetylglucosamine 1- phosphate transferase	Glycosaminoglycans (mucopolysaccharides), sphingolipids	Dysmorphic facial features, hepatosplenomegaly, muscle weakness and joint contractures	Inclusion Cells

Answers

	Tay- Sachs Disease: Gangliosides; Enzyme: Hexosaminidase deficiency
	Gaucher Disease; Glucoceroside; Glucocerebrosidase deficiency
	Niemann-Pick Disease; Sphingomyelin; Sphingomyelinase deficiency
	I-cell Disease; Impaired trafficking of lysosomal enzymes; Phosphotransferase deficiency (Defective Mannose 6-P phosphorylation in Golgi)

Complete the following table:

Disorder	Defective Enzyme	Accumulating substrate	Clinical Manifestations	Histological findings
Hurler Syndrome AR	L-iduronidase	Heparan Sulfate, Dermatan Sulfate	Corneal clouding, dysmorphic facial features, hepatosplenomegaly, joint contractures	
Hunter Syndrome XR	Iduronate sulfatase	Heparan Sulfate, Dermatan Sulfate	NO Corneal clouding, dysmorphic facial features, hepatosplenomegaly, Joint contractures, "Mongolian" spot on lower back	
Tay- Sachs Disease AR	Hexosaminidase A	Ganglioside (GM2)	Cherry-red spot on retina (blindness), neurological and developmental regression. No hepatosplenomegaly	Whorls/onion – shell appearance
Gaucher Disease AR	Glucosidase (Glucocerebrosidase)	Glucosyl-ceramide(Glucocerebroside)	Hepatosplenomegaly, bone pain. Spares the brain	Crumpled paper appearance
Fabry Disease XR	Galactosidase	Globoside (ceramide Trihexoside)	Numbness & tingling in extremities (acroparasthesia); red spots on buttocks and thighs (angiokeratoma) heart, kidney dysfunction	
Niemann- Pick Disease AR	Sphingomyelinase	Sphingomyelin	Type A- cherry red spot on retina. Type B- hepatosplenomegaly	Foamy cells
Metachromatic Leukodystrophy AR	Aryl Sulfatase A	Sulfatides	Progressive paralysis, motor & sensory abnormalities	
Pompe Disease (GSD type II) AR	Lysosomal acid maltase (1-4 glucosidase)	Glycogen	Muscle weakness, CVS, liver, kidney dysfunction	
I-Cell Disease AR	N-acetylglucosamine 1- phosphate transferase	Glycosaminoglycans (mucopolysaccharides), sphingolipids	Dysmorphic facial features, hepatosplenomegaly, muscle weakness and	Inclusion Cells

			joint contractures	
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Practice Questions for Metabolic Changes in Diabetes Mellitus **with explanations**

1. A 22-year-old woman comes to the physician because of a 2-week history of unintentional and progressive weight loss. She also says she has increased urine frequency and increased thirst. Her body mass index is 20kg/m². Laboratory studies show fasting plasma glucose is 130 mg/dL (70-100mg/dL). Urinalysis shows mild glucosuria, and ketonuria. Which of the following is most likely responsible for the clinical presentation in this patient?
 - A. Decreased gluconeogenesis
 - B. Increased secretion of insulin
 - C. Decreased proteolysis
 - D. Decreased number of GLUT-4
 - E. Activation of liver glycogenesis
2. A 14-year-old boy is brought to emergency department by his coach as he collapsed while playing cricket. Physical examination shows mild disorientation. He is hypotensive, tachycardic with rapid labored breathing. Laboratory studies show serum Na⁺ is 145 mmol/L (136-146); K⁺ is 5.3 mmol/L (3.5-5); and glucose is 521 mg/dL (70-100). Urinalysis show 4+ glucosuria and 4+ ketonuria levels, but no blood or nitrite. Which of the following enzymes is most likely active in this patient?
 - A. Hexokinase
 - B. Phosphofructokinase-1
 - C. Pyruvate kinase
 - D. Glycogen synthase
 - E. Pyruvate carboxylase
3. A 16-year-old boy is brought to the physician by his parents because of a 2-week history of fatigue and unintentional weight loss. He says he feels thirsty and passes urine frequently. He is hypotensive and has tachycardia. Laboratory studies show plasma glucose is 230 mg/dL (70-100). Urine dipstick tests show glycosuria and ketonuria. Which of the following additional laboratory findings is most likely in this patient?
 - A. Increased serum acetoacetate levels
 - B. Increased serum bicarbonate levels
 - C. Decreased anion gap
 - D. Low levels of urinary phosphate excretion
 - E. Low serum 3-hydroxybutyrate levels

4. A 60-year-old man comes to the physician because of a 2-month history of vision problems. He also has a nonhealing ulcer on his leg for 2 weeks. Physical examination shows a 3-cm ulcer that overlies the right first metatarsal head. There is diminished sensation to pinprick in his feet. Which of the following enzymes is most likely associated with his impaired vision?
- A. Transketolase
 - B. Transaldolase
 - C. Glycogen phosphorylase
 - D. Aldose reductase
 - E. Glycogen synthase
 - F. Lipoprotein lipase
5. A 25-year-old woman is brought to the emergency department because of a 3-hour history of confusion, abdominal pain and vomiting. Two days ago she had gastroenteritis. Physical examination shows dry mucus membranes, and a fruity odor of breath. Laboratory studies show very high plasma glucose and markedly reduced serum bicarbonate. Which of the following is the most likely diagnosis?
- A. Insulin resistance syndrome
 - B. Chronic pancreatitis
 - C. Type 1 diabetes mellitus
 - D. Type 2 diabetes mellitus
 - E. Addison disease
 - F. Cushing syndrome
 - G. Insulinoma
6. A 15-year-old girl is brought to the physician by her parents because of a 1-week history of progressive and unintentional weight loss. She was diagnosed with Type 1 diabetes one year earlier. Urine dipstick test shows glycosuria. Which of the following metabolic findings is most likely to be observed in this patient?
- A. Decreased ketogenesis
 - B. Low anion gap
 - C. Accelerated lipolysis
 - D. Reduced hormone sensitive lipase activity
 - E. Decreased proteolysis

Key: 1 **D**, 2 **E**, 3 **A**, 4 **D**, 5 **C**, 6 **C**.

Explanations and comments for: Metabolic changes in Diabetes mellitus

1. A 22-year-old woman comes to the physician because of a 2-week history of unintentional and progressive weight loss. She also says she has increased urine frequency and increased thirst. Her body mass index is 20kg/m². Laboratory studies show fasting plasma glucose is 130 mg/dL (70-100mg/dL). Urinalysis shows mild glucosuria, and ketonuria. Which of the following is most likely responsible for the clinical presentation in this patient?

- A. Decreased gluconeogenesis : Gluconeogenesis is increased
- B. Increased secretion of insulin : In type 1 diabetes mellitus, beta cells are destroyed leading to low insulin levels
- C. Decreased proteolysis : there is increased breakdown of protein due to low insulin levels
- D. **Decreased number of GLUT-4 correct answer:** there is reduced number of GLUT 4 due to low insulin levels
- E. Activation of liver glycogenesis - there is reduced activity of liver glycogen synthase due to low insulin levels

2. A 14-year-old boy is brought to emergency department by his coach as he collapsed while playing cricket. Physical examination shows mild disorientation. He is hypotensive, tachycardic with rapid labored breathing. Laboratory studies show serum Na⁺ is 145 mmol/L (136-146); K⁺ is 5.3 mmol/L (3.5-5); and glucose is 521 mg/dL (70-100). Urinalysis show 4+ glucosuria and 4+ ketonuria levels, but no blood or nitrite. Which of the following enzymes is most likely active in this patient?

- A. Hexokinase – Activity in peripheral tissues is not affected by insulin
- B. Phosphofructokinase-1
- C. Pyruvate kinase
- D. Glycogen synthase – Activity is reduced due to low insulin levels
- E. **Pyruvate carboxylase: Correct answer:** Gluconeogenesis is enhanced in this condition. This is one of the enzymes in this pathway. Options A to C are irreversible steps in the glycolytic pathway.

3. A 16-year-old boy is brought to the physician by his parents because of a 2-week history of fatigue and unintentional weight loss. He says he feels thirsty and passes urine frequently. He is hypotensive and has tachycardia. Laboratory studies show plasma glucose is 230 mg/dL (70-100). Urine dipstick tests show glycosuria and ketonuria. Which of the following additional laboratory findings is most likely in this patient?

- A. **High levels of acetoacetate: Correct answer:** There is increased production of ketone bodies due to excessive lipolysis and they are used as alternative sources of energy
- B. Increased serum bicarbonate levels Bicarbonate serves as a buffer and is lost in these patients due to buffering the ketone bodies (weak acids). Bicarbonate levels are low in diabetic ketoacidosis.

- C. Decreased anion gap : There is an increase in unmeasured anions (ketone bodies) and there is increased anion gap metabolic acidosis
- D. Low levels of urinary phosphate excretion : Phosphate and ammonia serve as renal tubular buffers. In diabetic ketoacidosis, there is increased phosphate and ammonium excretion by the kidney.
- E. Low serum 3-hydroxybutyrate levels : There is increased ketogenesis due to uncontrolled lipolysis in the adipose tissue

4. A 60-year-old man comes to the physician because of a 2-month history of vision problems. He also has a nonhealing ulcer on his leg for 2 weeks. Physical examination shows a 3-cm ulcer that overlies the right first metatarsal head. There is diminished sensation to pinprick in his feet. Which of the following enzymes is most likely associated with his impaired vision?

- A. Transketolase : Is a key enzyme in the non-oxidative part of pentose phosphate pathway. It requires TPP
- B. Transaldolase : Is an enzyme in the non-oxidative pathway of pentose phosphate pathway
- C. Glycogen phosphorylase : catalyzes the rate-limiting step in glycogenolysis and is activated by low insulin: glucagon ratio
- D. **Aldose reductase: Correct answer: Catalyzes the reduction of glucose to sorbitol which accumulates in the lens leading to impaired vision**
- E. Glycogen synthase : converts glucose to glycogen and is more active in the fed state
- F. Lipoprotein lipase : it breaks down triglycerides (TAG) in VLDL and chylomicrons, producing free fatty acids which are used for energy in muscle or stored as adipose TAG. In diabetics, low activity of lipoprotein lipase results in elevated serum TAG and is responsible for the dyslipidemia and macrovascular complications

5. A 25-year-old woman is brought to the emergency department because of a 3-hour history of confusion, abdominal pain and vomiting. Two days ago she had gastroenteritis. Physical examination shows dry mucus membranes, and a fruity odor of breath. Laboratory studies show very high plasma glucose and markedly reduced serum bicarbonate. Which of the following is the most likely diagnosis?

- A. Insulin resistance syndrome – Insulin resistance is generally a risk factor for type 2 diabetes mellitus
- B. Chronic pancreatitis : inflammation of the pancreas over time. Presents as malabsorption and steatorrhea
- C. **Type 1 diabetes mellitus: Correct answer: Patients with type 1 DM can present with diabetic ketoacidosis. Features of ketoacidosis are usually observed such as fruity breath odor, metabolic acidosis and hyperglycemia.**
- D. Type 2 diabetes mellitus : Patients usually present as hyperosmolar hyperglycemic non-ketotic state. Ketone body levels are usually not elevated in these patients.
- E. Addison disease : due to adrenal insufficiency. Ketoacidosis is not observed in this condition. Patients have hypoglycemia, low sodium and high potassium levels.

- F. Cushing syndrome : due to prolonged exposure to cortisol. Ketoacidosis is not a feature of this condition. These patients have hyperglycemia, hypertension and abdominal striae.
- G. Insulinoma : tumor of the pancreas. Causes hypoglycemia due to increased production of insulin.

6. A 15-year-old girl is brought to the physician by her parents because of a 1-week history of progressive and unintentional weight loss. She was diagnosed with Type 1 diabetes one year earlier. Urine dipstick test shows glycosuria. Which of the following metabolic findings is most likely to be observed in this patient?

- A. Decreased ketogenesis : increased production of ketone bodies is usually found in uncontrolled type 1 diabetics
- B. Low anion gap : increased production of ketone bodies (unmeasured anions) results in high anion gap metabolic acidosis
- C. **Accelerated lipolysis: Correct answer: breakdown of adipose tissue TAG is enhanced to release fatty acids due to uncontrolled activity of hormone sensitive lipase**
- D. Reduced hormone sensitive lipase activity : increased activity of HS lipase due to lack of insulin.
- E. Decreased proteolysis : In uncontrolled diabetics there is increased breakdown of proteins to produce glucogenic amino acids that can be converted to glucose in the liver via gluconeogenesis

Nutrition and intensive care: Practice Questions

1. A 60-year-old man is brought to the physician because of a 1-day history of feeling weak and tired. He is in a refugee camp for 2 months and has had no food for the past 15 days. He has no fever or infection. Which of these findings is most likely observed in this patient?

- A. Increased plasma glucose levels
- B. Increased plasma ketone bodies
- C. Low plasma free fatty acids
- D. High acute phase reactants
- E. Low fatty acid oxidation in liver
- F. High resting energy expenditure

2. A 74-year-old man is brought to the emergency department because of a 6-hour history of lethargy. He has a 15-year history of type 2 diabetes mellitus and is on treatment with oral hypoglycemia agents. He has had fever due to a respiratory infection for the past 2 days. Laboratory studies on admission show plasma glucose level is 680mg/dL. Which of the following findings is most likely?

- A. Increased muscle protein synthesis
- B. Reduced urinary nitrogen loss
- C. Low C-reactive protein levels
- D. Low fatty acid oxidation in liver

- E. Low metabolic rate
- F. Hyperosmolarity

3. A 14-year-old girl is brought to the emergency department by her parents in an unconscious state. She has a 2-year history of type 1 diabetes mellitus. She has had a 3-day history of fever with productive cough. Laboratory studies show plasma glucose level is 380mg/dL (70-100mg/dL). Which of the following additional findings is most likely in this patient?

- A. Increased plasma ketone bodies
- B. Low activity of hormone sensitive lipase
- C. Increased GLUT-4 in muscle
- D. Reduced gluconeogenesis in liver
- E. Positive nitrogen balance
- F. Increased arterial pH

4. A 45-year-old woman is brought to the emergency department because of a 6-hour history of abdominal pain. She has had a 2-day history of fever. Her temperature is 39°C (102.2°F). Physical examination shows abdominal guarding. She is diagnosed with peritonitis and antibiotics are prescribed. It is observed that she is not responding to the treatment. Which of the following findings would most likely be present 5 days later?

- A. Low resting metabolic rate
- B. High C-reactive protein levels
- C. Increased lipogenesis in adipose tissue
- D. Increased muscle protein synthesis
- E. Decreased urea excretion in urine
- F. Decreased hepatic gluconeogenesis
- G. Positive nitrogen balance

5. A 45-year-old woman is brought to the emergency department because of a 6-hour history of abdominal pain. She has had a 2-day history of fever. Her temperature is 39°C (102.2°F). Physical examination shows abdominal guarding. She is diagnosed with peritonitis and antibiotics are prescribed. She starts improving in 10 days. Which of the following findings would most likely be present 3 weeks later (compared to the day of admission)?

- A. Higher resting metabolic rate
- B. Higher C-reactive protein levels
- C. Higher rates of adipose tissue lipolysis
- D. Higher rates of muscle protein synthesis
- E. Higher urea excretion in urine
- F. Higher epinephrine levels
- G. Higher circulating PGE2 levels

Starvation vs Critical illness: Fill-in the table

	Simple starvation	Critical illness
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BMR (metabolic rate)		
Presence of inflammatory mediators		
Major fuel oxidized		
Ketone body levels		
Negative N2 balance and hepatic urea formation		
Blood glucose level		
Gluconeogenesis & muscle proteolysis		
Hepatic protein synthesis		
Circulating insulin levels		

Nutrition and intensive care: Practice Questions

Answer Key:

1. B

2. F

3. A

4. B

5. D

Nutrition and intensive care: Practice Questions : Brief explanations

1. A 60-year-old man is brought to the physician because of a 1-day history of feeling weak and tired. He is in a refugee camp for 2 months and has had no food for the past 15 days. He has no fever or infection. Which of these findings is most likely observed in this patient?

Comment: The person is in uncomplicated starvation

- A. Increased plasma glucose levels – Plasma glucose levels are low (lower limit of normal) during starvation
- B. Increased plasma ketone bodies – Correct answer: Ketosis occurs in prolonged starvation to provide an alternate fuel for the brain; Ketosis also has a protein sparing effect**
- C. Low plasma free fatty acids – HS lipase in adipose tissue is active and free fatty acid levels are increased
- D. High acute phase reactants – In uncomplicated starvation, acute phase proteins are not increased
- E. Low fatty acid oxidation in liver – Beta oxidation is increased in liver; Acetyl CoA activates gluconeogenesis and is used for ketogenesis in liver
- F. High resting energy expenditure – There is an adaptive decrease in resting metabolic rate in starvation (Prolongs survival)

2. A 74-year-old man is brought to the emergency department because of a 6-hour history of lethargy. He has a 15-year history of type 2 diabetes mellitus and is on treatment with oral hypoglycemia agents. He has had fever due to a respiratory infection for the past 2 days. Laboratory studies on admission show plasma glucose level is 680mg/dL. Which of the following findings is most likely?

- A. Increased muscle protein synthesis – Muscle protein breakdown is enhanced in the flow state
- B. Reduced urinary nitrogen loss – Urinary urea losses are increased and indicate the degree of proteolysis
- C. Low C-reactive protein levels – CRP levels are increased as it is an acute phase protein
- D. Low fatty acid oxidation in liver – There is increased beta oxidation in liver due to hypermetabolism
- E. Low metabolic rate – Metabolic rate and oxygen utilization are increased
- F. Hyperosmolarity – Correct answer: may be seen in a type 2 diabetic following an infection; Hyperosmolar hyperglycemic state is a complication of type 2 diabetes following an infection**

3. A 14-year-old girl is brought to the emergency department by her parents in an unconscious state. She has a 2-year history of type 1 diabetes mellitus. She has had a 3-day history of fever with productive cough. Laboratory studies show plasma glucose level is 380mg/dL (70-100mg/dL). Which of the following additional findings is most likely in this patient?

Comment: The patient is a type 1 diabetic with a high risk of ketoacidosis following an infection

- A. Increased plasma ketone bodies – Correct answer: Ketoacidosis is a common complication in type 1 diabetics following an infection or stress**
- B. Low activity of hormone sensitive lipase – HS Lipase is very active due to low insulin levels and the additional insulin resistance. Insulin requirements are higher during an incidental infection.
- C. Increased GLUT-4 in muscle – There are fewer GLUT-4 due to low insulin and insulin resistance
- D. Reduced gluconeogenesis in liver – Increased liver gluconeogenesis is responsible for the elevated plasma glucose levels
- E. Positive nitrogen balance – There is negative nitrogen balance following an infection

- F. Increased arterial pH – the arterial pH is most likely decreased (metabolic acidosis) with low HCO_3^- (loss by buffering) and low PCO_2 (compensatory hyperventilation)

4. A 45-year-old woman is brought to the emergency department because of a 6-hour history of abdominal pain. She has had a 2-day history of fever. Her temperature is 39°C (102.2°F). Physical examination shows abdominal guarding. She is diagnosed with peritonitis and antibiotics are prescribed. It is observed that she is not responding to the treatment. Which of the following findings would most likely be present 5 days later?

Comment: The patient is in the 'flow phase' of critical illness

- A. Low resting metabolic rate – Metabolic rate and oxygen consumption are higher
- B. High C-reactive protein levels – Correct answer: CRP levels are elevated as the inflammatory process is active. CRP is an acute phase protein**
- C. Increased lipogenesis in adipose tissue – During flow phase there is increased lipolysis, due to activation of hormone sensitive lipase
- D. Increased muscle protein synthesis – There is increased muscle proteolysis and amino acids are used for gluconeogenesis
- E. Decreased urea excretion in urine – Urea excretion in urine is increased and is an indicator of extent of proteolysis
- F. Decreased hepatic gluconeogenesis – Gluconeogenesis in the liver is active due to cortisol, glucagon and epinephrine and insulin resistance. As a result, plasma glucose levels are elevated.
- G. Positive nitrogen balance – There is negative nitrogen balance in the flow phase

5. A 45-year-old woman is brought to the emergency department because of a 6-hour history of abdominal pain. She has had a 2-day history of fever. Her temperature is 39°C (102.2°F). Physical examination shows abdominal guarding. She is diagnosed with peritonitis and antibiotics are prescribed. She starts improving in 10 days. Which of the following findings would most likely be present 3 weeks later (compared to the day of admission)?

Comment: The patient will be in the recovery 'anabolic phase' after 3 weeks

- A. Higher resting metabolic rate – there is normalization of metabolic rate
- B. Higher C-reactive protein levels – Normalization of C-reactive protein levels
- C. Higher rates of adipose tissue lipolysis – Increased adipose tissue lipogenesis is observed in the anabolic phase
- D. Higher rates of muscle protein synthesis – Correct answer: Increased tissue protein synthesis is seen in the anabolic phase and dietary protein requirements are higher**
- E. Higher urea excretion in urine – There is decreased urea excretion in urine and the patient will most likely be in positive nitrogen balance
- F. Higher epinephrine levels – Normalization of epinephrine levels in recovery phase
- G. Higher circulating PGE_2 levels – As the inflammation resolves and during recovery phase, there is lowering of PGE_2 levels

Starvation vs Critical illness: Answers

	Simple starvation	Critical illness
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BMR (metabolic rate)	↓ (Hypometabolic)	↑ (Hypermetabolic)
Presence of inflammatory mediators	—	+++
Major fuel oxidized	Fat	Protein and fat
Ketone body levels	+++	± (does not correspond to the increase in free fatty acids)
Negative N ₂ balance and hepatic urea formation	+	+++
Blood glucose level	Slightly lower	Increased (insulin resistance)
Gluconeogenesis & muscle proteolysis	+	+++
Hepatic protein synthesis	+ (Basal)	+++ (acute phase proteins)
Circulating insulin levels	Low	+++ (insulin resistance)

Comparison of Critical illness (flow phase) and prolonged starvation

Comparison of Carbohydrate metabolism

	Critical illness (Flow phase)	Prolonged starvation
Blood glucose level	Increased	Decreased
Insulin levels	High and insulin resistance	Low
Energy for the Brain	Glucose	Glucose + Ketone bodies
Glucose uptake by adipose and muscle (GLUT4)	Low due to insulin resistance	Low due to low levels of insulin

Gluconeogenesis	Highly active to maintain the high blood glucose levels	Active to maintain the blood glucose levels
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Comparison of Lipid metabolism

	Critical illness (Flow phase)	Prolonged starvation
Hormone sensitive lipase	Phosphorylated and very active due to epinephrine and cortisol	Phosphorylated and active due to low levels of insulin
Lipolysis	Very active; Increases free fatty acid levels in circulation	Active; Increases free fatty acid levels in circulation
Insulin levels	High and insulin resistance	Low
Energy for brain	Glucose	Glucose + Ketone bodies
Ketone body levels in circulation	LOW - Increased ketone body utilization by peripheral tissues (due to hypermetabolism)	Elevated

Comparison of protein metabolism

	Critical illness (Flow phase)	Prolonged starvation
Muscle protein	Muscle proteolysis >>> Muscle protein synthesis	Muscle proteolysis> Muscle protein synthesis
Proteolysis (protein catabolism)	Very active – results in muscle wasting	Proteolysis (protein catabolism) – active – Muscle wasting not as pronounced
Negative nitrogen balance	Negative nitrogen balance is marked and dependent on severity of illness	Negative nitrogen balance (not as marked) – due to protein sparing effect of ketone bodies in prolonged starvation
Urinary Urea	Urinary urea (Urinary nitrogen) excretion is elevated – estimates the degree of proteolysis	Urinary urea excretion is reduced (Protein sparing effect of ketone bodies)